



MONITORING THE EXPECTED ALCOHOL DEGREE IN ORDER TO PREDICT GRAPES HARVEST
BY

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SUBMITTED
TO



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30TH APRIL, 2026

1.0 Backgrounds of the Practical

Viticulture in the Douro region requires precise management due to substantial spatial heterogeneity in topography, solar exposure, soil conditions, and vine vigor. Vineyard productivity is influenced not only by grape variety and management practices but also by temporal and spatial variation in sugar accumulation. Monitoring grape maturation is therefore essential for predicting harvest timing and ensuring desired wine quality. At Quinta de Nossa Senhora de Lurdes, vineyard monitoring combines conventional field methods with precision agriculture technologies. The central objective is to estimate expected alcohol degree by measuring Brix values, where 1 °Bx approximates 0.55 ° alcohol. Thus, grapes reaching 25.5 °Bx are expected to produce roughly 14% alcohol, an indicator of suitable harvest maturity. This report expands vineyard monitoring beyond traditional sampling by incorporating GIS cartography, geostatistics, and remote sensing, thereby converting point-based measurements into management intelligence across the entire property ([Aranha, 2026](#)).

Quinta de Nossa Senhora de Lurdes (Qt. N. S. de Lurdes), located in Vila Real within the Douro viticultural context, serves as an advanced study site for vineyard monitoring using traditional field sampling, Geographic Information Systems (GIS), geostatistics, and remote sensing technologies. This report evaluates vineyard productivity through systematic Brix monitoring to estimate grape sugar accumulation, forecast alcohol potential, and determine optimal harvest timing. The study integrates refractometer measurements, regression analysis, thematic cartography, Thiessen polygons, Inverse Distance Weighted (IDW) interpolation, drone-assisted remote sensing, and NDVI-based vegetation assessment. Traditional sampling demonstrated progressive Brix increases from July to late August, with average values rising from 9.19 °Bx on July 15 to 20.54 °Bx on August 30. Regression modeling projected harvest suitability at approximately 25.5–27 °Bx around Julian day 256 (September 13), corresponding to wine alcohol potential near 14%. GIS-based approaches transformed discrete point measurements into spatially explicit productivity zones, allowing parcel prioritization for selective harvesting. NDVI modeling further supported predictive vineyard management by correlating spectral reflectance with sugar concentration. The integration of GNSS, QField, QGIS, drones, and geostatistical models demonstrates a precision agriculture framework capable of optimizing harvest timing, improving grape quality, reducing operational uncertainty, and enhancing site-specific management for irrigation, fertilization, and selective production.

2.0 Study Area Characterization

The study area is UTAD's vineyard, Quinta de Nossa Senhora de Lurdes, located in Vila Real, Portugal. According to the practical and theoretical modules, the vineyard covers approximately 6 hectares and is georeferenced within WGS84 UTM Zone 29N coordinates ([Aranha, 2026](#)):

- Lower left corner: X = 605620; Y = 4571120
- Upper right corner: X = 605960; Y = 4571400
- The property is situated within a Mediterra
- near climatic influence zone characterized by warm summers, variable slopes, and strong topographic controls on grape maturation. Morphological complexity significantly influences vine water stress, sunlight interception, and berry sugar concentration.

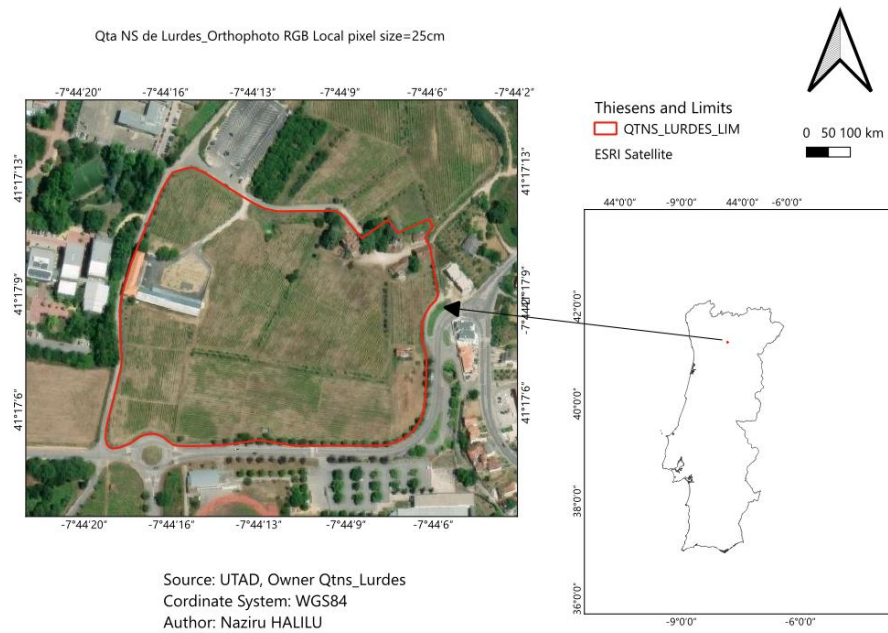


Figure 1: Location of Farm Area in RGB

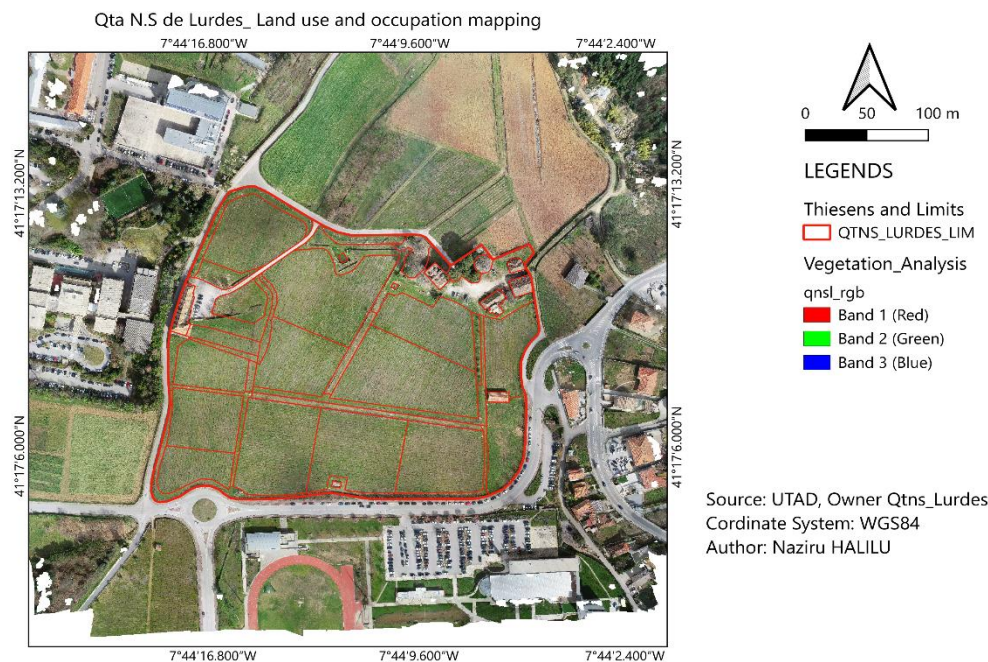


Figure 2: Qta N.S de Lurdes Land Use and Occupation Mapping

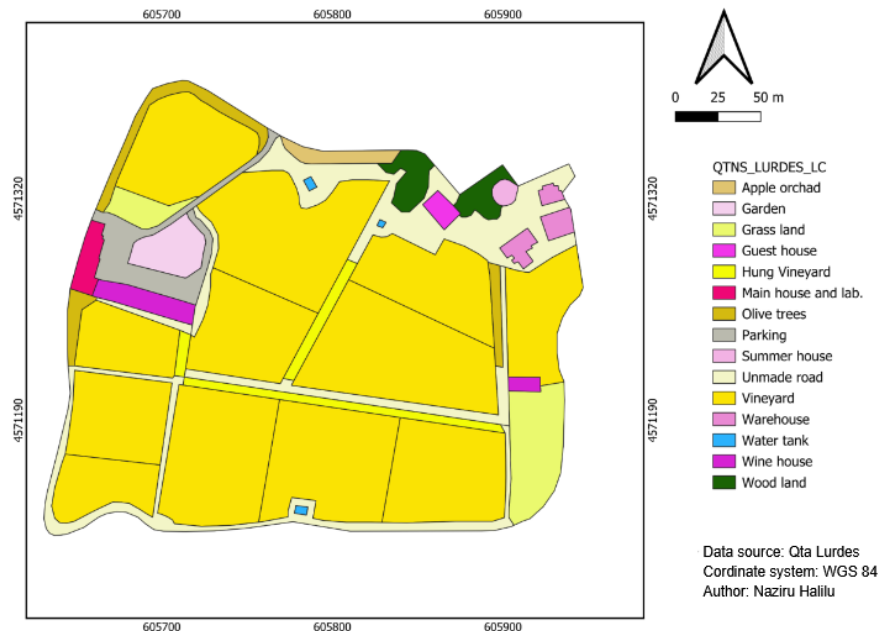


Figure 3 : N.S Land use Mapping

Table1 : Land Use and Land Cover Classification with Parcel Count and Area Distribution

LAND COVER	Agricultural Parcels	Agricultural Area (m ²)	Human Parcels	Human Area (m ²)	Forestry Parcels	Forestry Area (m ²)	Water Parcels	Water Area (m ²)	Total Parcels	Total Area (m ²)
Apple orchard	1	521.62	0	0	0	0	0	0	1	521.62
Garden	0	0	1	1063.84	0	0	0	0	1	1063.84
Guest house	0	0	1	248.94	0	0	0	0	1	248.94
Grass land	2	2833.63	0	0	0	0	0	0	2	2833.63
Main house and lab	0	0	1	494.85	0	0	0	0	1	494.85
Olive trees	3	1565.52	0	0	0	0	0	0	3	1565.52
Parking	0	0	1	1883.51	0	0	0	0	1	1883.51
Summer house	0	0	1	181.01	0	0	0	0	1	181.01
Unmade road	0	0	4	8663.54	0	0	0	0	4	8663.54
Warehouse	0	0	3	644.30	0	0	0	0	3	644.30
Water tank	0	0	0	0	0	0	3	83.94	3	83.94
Wood land	0	0	0	0	2	1169.91	0	0	2	1169.91
Wine house	0	0	2	859.70	0	0	0	0	2	859.70
Vineyard	12	43356.69	0	0	0	0	0	0	12	43356.69
Hung Vineyard	3	1547.42	0	0	0	0	0	0	3	1547.42
Grand Total	21	49824.88	14	14039.69	2	1169.91	3	83.94	40	65118.42

The pivot analysis clearly demonstrates that Quinta de Nossa Senhora de Lurdes is fundamentally an agriculturally driven estate, with agriculture accounting for over 72% of the total mapped land area. Within this agricultural structure, viticulture is overwhelmingly dominant, as vineyards and hung vineyards together occupy more than 87% of all agricultural land. This confirms that grape production is the estate's principal productive function and economic foundation. Among vineyard varieties, Touriga Nacional is the most spatially dominant cultivar, covering approximately 38% of the vineyard area, reinforcing its importance as the flagship red grape variety of the estate. This is consistent with the strategic role of Touriga Nacional in premium Douro wine production due to its strong enological value. Tinta Roriz also represents a major varietal component, while white varieties such as Viozinho, Arinto, C3dega de Larinho, and Rabigato collectively contribute to varietal diversity, allowing the estate to maintain both red and white production potential. Human land use occupies nearly 23% of the total area, which is relatively significant and indicates a well-developed operational infrastructure. The presence of wine houses, warehouses, roads, laboratories, guest houses, and residential structures suggests that the estate is not solely a cultivation site but an integrated production and management complex. In particular, the extensive unmade road network highlights the logistical importance of internal accessibility for machinery, harvesting, and estate operations. Forestry and water occupy relatively small proportions of the total estate but remain strategically important. Woodland contributes to ecological stability and biodiversity, while water tanks support irrigation and water security critical factors in precision viticulture, especially under Mediterranean climatic pressures.

Overall, the pivot table reveals a highly specialized viticultural estate supported by diversified infrastructure and complementary land uses. The dominance of vineyards, especially premium red cultivars, positions the property strongly for wine production, while the presence of white varieties and auxiliary agricultural land enhances resilience and production flexibility. This land structure reflects a balanced model of productive efficiency, operational functionality, and environmental integration.

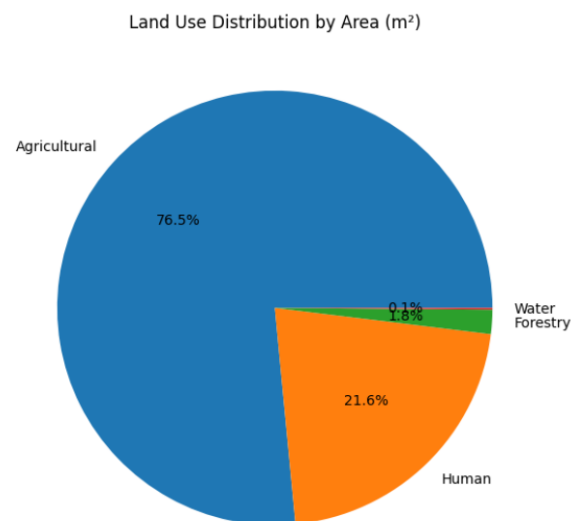


Figure 4 Land Use Distribution by Area

2.1 Topographic Importance

Slope and aspect are among the most influential topographic factors controlling vineyard performance because they directly shape the environmental conditions under which vines grow, mature, and ultimately determine grape quality. In viticultural landscapes such as Quinta de Nossa Senhora de Lurdes, where terrain heterogeneity is pronounced, variations in slope gradient and directional exposure create significant differences in microclimate, soil behavior, and vine physiology across relatively short distances. These variations make topographic analysis an essential component of precision vineyard management ([Aranha, 2026](#)). Slope primarily affects the movement of water, solar energy distribution, and soil stability. Steeper slopes generally promote rapid surface runoff, reducing water retention and increasing drainage efficiency. While this can be beneficial in preventing waterlogging and root diseases, excessive slope may also accelerate soil erosion and reduce moisture availability during dry periods. Conversely, flatter areas tend to retain more water, which may support greater vegetative growth but can also create risks of poor drainage, compaction, or excessive vigor that may dilute grape quality. Therefore, understanding slope gradients allows vineyard managers to balance irrigation, erosion control, and cultivation practices according to local terrain conditions. Aspect, or the directional orientation of slopes, significantly influences the amount and intensity of solar radiation intercepted by vineyard parcels. South-facing slopes in the Northern Hemisphere generally receive more direct sunlight throughout the growing season, resulting in higher heat accumulation, enhanced photosynthetic activity, and faster grape maturation. These parcels often reach optimal sugar concentration earlier and may produce grapes with higher Brix values and alcohol potential. In contrast, north-facing or shaded slopes typically receive less solar exposure, leading to cooler conditions, slower sugar accumulation, and delayed maturation. East- and west-facing slopes may also differ, with eastern exposures often benefiting from gentler morning sunlight while western slopes may experience intense afternoon heat, potentially increasing evapotranspiration and heat stress ([Aranha, 2026](#)). Temperature accumulation is closely tied to both slope and aspect. Elevated and sun-exposed sites generally experience distinct thermal regimes compared to valley bottoms or shaded depressions. Temperature influences vine phenology, including budburst, flowering, veraison, and harvest timing. Warmer parcels may accelerate ripening, while cooler zones may preserve acidity and aromatic complexity. This spatial thermal variability is particularly important in wine production because different maturation rates can affect both grape composition and desired wine style. Topography also impacts soil formation and depth. Steep upper slopes often have shallower soils due to erosion and reduced organic matter accumulation, which can limit vine vigor but sometimes improve fruit concentration through moderate stress. Lower slopes or depositional zones may possess deeper, more fertile soils with greater water-holding capacity, supporting stronger vegetative growth but potentially requiring canopy management to maintain fruit quality. Thus, soil depth variation caused by terrain influences root development, nutrient access, and drought resilience. Because slope and aspect collectively determine solar interception, drainage, soil characteristics, and temperature patterns, they strongly influence maturation speed across vineyard parcels. Higher elevations and south-facing slopes may ripen earlier and produce premium-quality grapes, while lower or shaded parcels may require adjusted harvest schedules or differentiated management strategies. GIS-based topographic analysis is therefore fundamental for management zoning, as it enables vineyard managers to spatially identify these variations and implement site-specific decisions such as selective harvesting, irrigation zoning, erosion mitigation, and varietal placement. By integrating slope and aspect data into

vineyard planning, producers can optimize grape quality, improve operational efficiency, and enhance the overall sustainability of viticultural systems.

3. Traditional Vineyard Monitoring Methodology

Traditional vineyard monitoring involves field technicians manually collecting grape berries across multiple sampling plots, extracting juice, and measuring sugar concentration using a refractometer.

3.1 Refractometer Principle

Degrees Brix (°Bx) represent grams of sucrose per 100 grams of aqueous solution and estimate sugar concentration. In viticulture:

$$1\text{ }^{\circ}\text{Bx} \approx 0.55\text{ }^{\circ}\text{Alcohol} \text{ (Aranha, 2026)} \dots\dots\dots(1)$$

Thus:

- 24 °Bx $\approx$ 13.2% alcohol
- 25.5 °Bx $\approx$ 14.0% alcohol

3.2 Sampling Timeline

Sampling was conducted systematically across five carefully selected dates during the grape maturation period: July 15, July 30, August 6, August 15, and August 30. These dates were strategically distributed throughout the growing season to capture the progressive biochemical and physiological changes occurring within the grape berries as they transitioned from early development through veraison and toward full ripening. By spacing sampling intervals over approximately two-week and one-week periods, the monitoring program provided a structured temporal framework for assessing sugar accumulation, maturation trends, and the vineyard's evolving harvest potential.

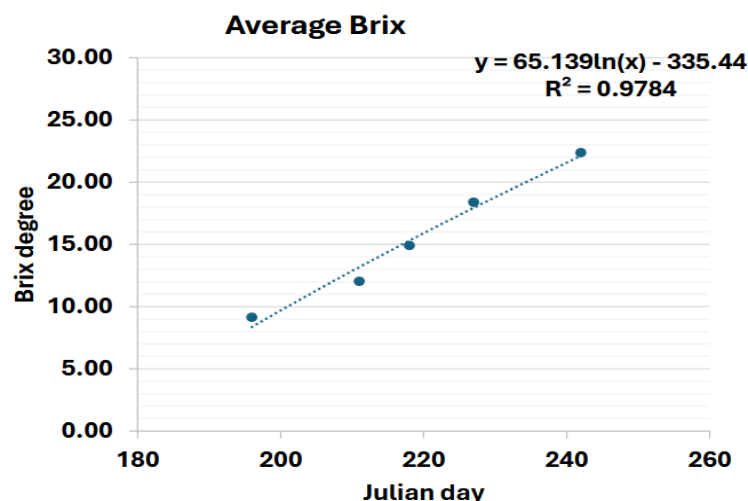


Figure 5: Adjusting the Regression Model

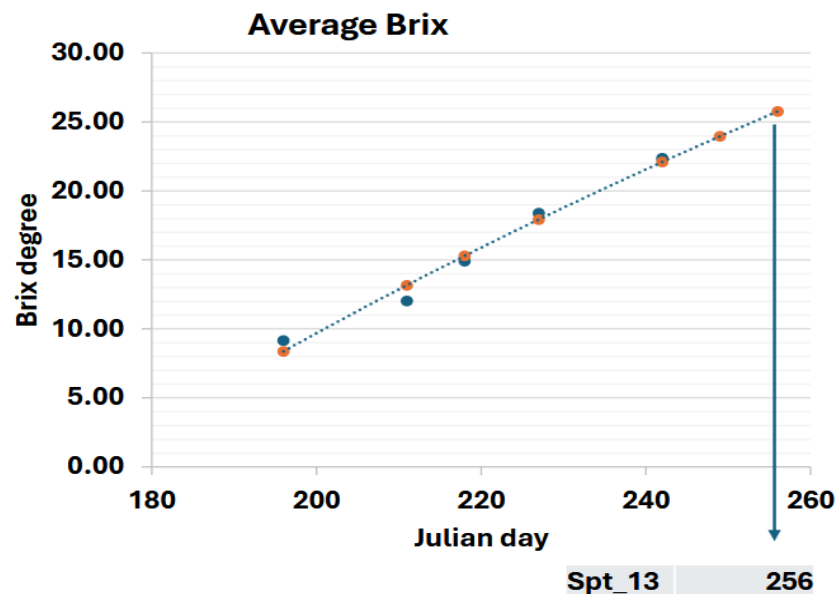


Figure 6: Forecasting the harvesting date Potentials

The first sampling date, July 15, represented the early phase of maturity assessment, when grape berries were still in the initial stages of sugar accumulation. At this point, Brix values were relatively low, reflecting the pre-veraison or early veraison stage, where berries are still developing and sugar concentration remains limited. This baseline measurement was essential for establishing the starting point of the maturation curve and provided a reference against which subsequent increases in sugar concentration could be compared.

The second sampling event on July 30 occurred approximately two weeks later and allowed vineyard managers to evaluate the onset of more rapid sugar accumulation. During this stage, vines generally experience intensified photosynthetic activity, and grapes begin transitioning more noticeably into ripening. Monitoring at this interval was particularly valuable for detecting the pace of maturation and identifying whether the vineyard was progressing according to expected seasonal patterns.

Sampling on August 6 provided a mid-season checkpoint, capturing the vineyard during a critical period when sugar accumulation often accelerates significantly. This date was important because it offered insights into the consistency of maturation progression and allowed for the early detection of spatial or temporal anomalies between vineyard parcels. Any areas lagging in development or showing excessive acceleration could be identified and addressed through targeted management interventions. The fourth sampling date, August 15, marked a more advanced ripening stage, when grapes typically approach commercially relevant sugar concentrations. Measurements taken during this period were particularly useful for refining harvest forecasts, as Brix values begin to provide clearer indications of final alcohol potential. This stage also helps vineyard managers assess the uniformity of maturation across the property, which is critical for selective harvesting strategies. The final sampling on August 30 served as a late-season maturity assessment, capturing grape conditions close to the anticipated harvest window. By this date, grapes had accumulated substantial sugar concentrations, and Brix values offered a strong basis for predicting the optimal harvest date through regression analysis. This final sampling was essential for validating maturation trends observed in earlier stages and for making practical decisions regarding labor scheduling, winery preparation, and parcel prioritization. Conducting sampling across these five dates created a comprehensive

temporal dataset that enabled the accurate monitoring of vineyard maturation dynamics. This repeated-measures approach not only improved understanding of sugar accumulation trends but also supported statistical modeling, GIS mapping, and precision viticulture applications. By observing the vineyard at multiple stages rather than relying on isolated measurements, managers gained a far more reliable basis for forecasting harvest readiness, optimizing grape quality, and implementing site-specific vineyard management practices.

3.3 Average Brix Progression

The progression of average Brix values across the monitoring period provides a clear representation of grape maturation dynamics within the vineyard and serves as one of the most important indicators for harvest forecasting. Brix measurements, which quantify the concentration of soluble sugars in grape juice, increased consistently throughout the sampling season, demonstrating the natural physiological process of sugar accumulation as grapes advanced from early development through veraison and into ripening. This progression reflects the combined effects of photosynthesis, carbohydrate translocation, climatic conditions, and vineyard-specific environmental factors such as slope, aspect, and vine vigor.

On July 15, the vineyard recorded an average Brix value of 9.19 °Bx, representing the early stage of grape development when berries had accumulated only moderate sugar concentrations. At this stage, the grapes were still physiologically immature, and much of the vine's energy was directed toward berry growth rather than sugar concentration. Such values are typical of pre-veraison or early veraison stages, where berries are transitioning from hard green fruit into softer, sugar-accumulating tissues ([Aranha, 2026](#)).

By July 30, average Brix values had risen to 12.15 °Bx, indicating that sugar accumulation was progressing steadily. This increase suggests that the vineyard had entered a more active maturation phase, where photosynthetic products were increasingly being stored in the berries. The rise of nearly 3 °Bx within this period reflects favorable environmental conditions and healthy vine function.

On August 6, the average Brix value reached 15.06 °Bx, marking a substantial acceleration in maturation. This stage often corresponds with active veraison, during which grape berries soften, change color, and rapidly accumulate sugars. The increase in Brix during this period is particularly important because it signals the beginning of commercially significant ripening.

By August 15, average Brix had climbed to 18.51 °Bx, confirming that grapes were moving deeper into the ripening stage. At this point, berry composition begins approaching levels relevant for wine production, and vineyard managers can more confidently estimate final harvest windows. The rate of sugar accumulation during this interval demonstrates efficient physiological maturation and likely reflects favorable sunlight exposure and temperature conditions.

Finally, on August 30, the vineyard achieved an average Brix value of 20.54 °Bx. This value indicates that the vineyard was approaching harvest maturity, though additional sugar accumulation was still required to achieve premium alcohol targets. The cumulative trend from 9.19 °Bx to 20.54 °Bx over approximately six weeks highlights a strong maturation curve and confirms that sugar accumulation accelerated as grapes progressed from veraison into advanced ripening. These values demonstrate a progressive and biologically consistent maturation trend characterized by increasing sugar concentration over time. The pattern is especially significant because it provides the quantitative foundation for predictive harvest modeling, allowing vineyard managers to estimate when grapes will achieve target alcohol potential and optimal oenological quality.

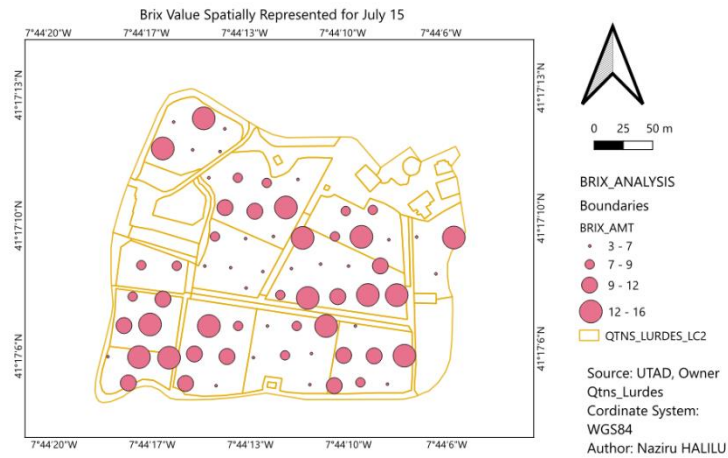


Figure 7: Brix Value Spatially Represented for July 15

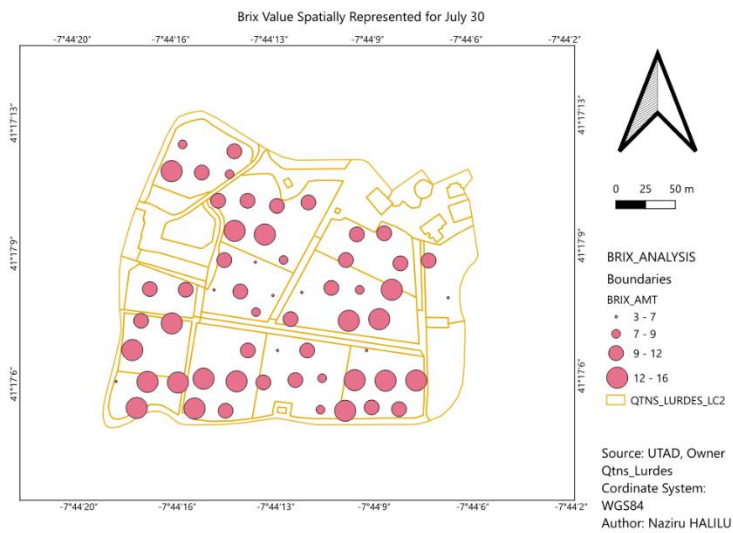


Figure 8: Brix Value spatially represented for July 30

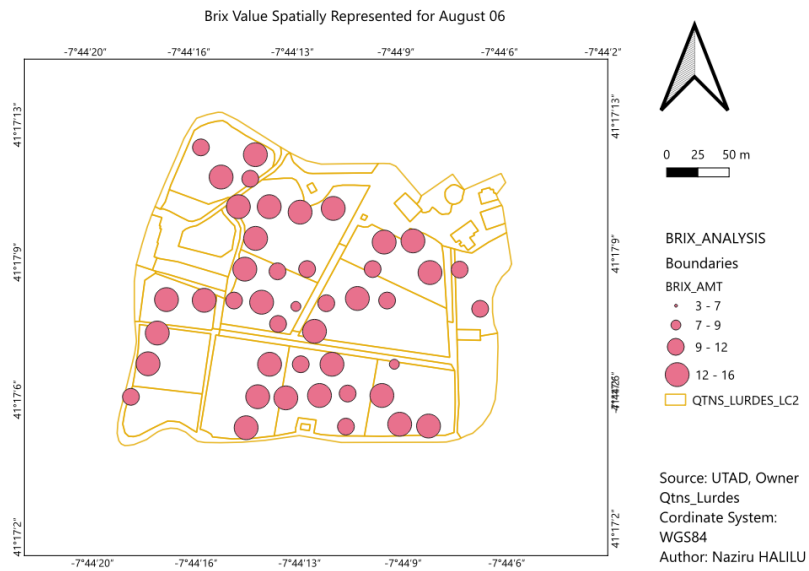


Figure 9: Brix Value spatially represented for Aug 06

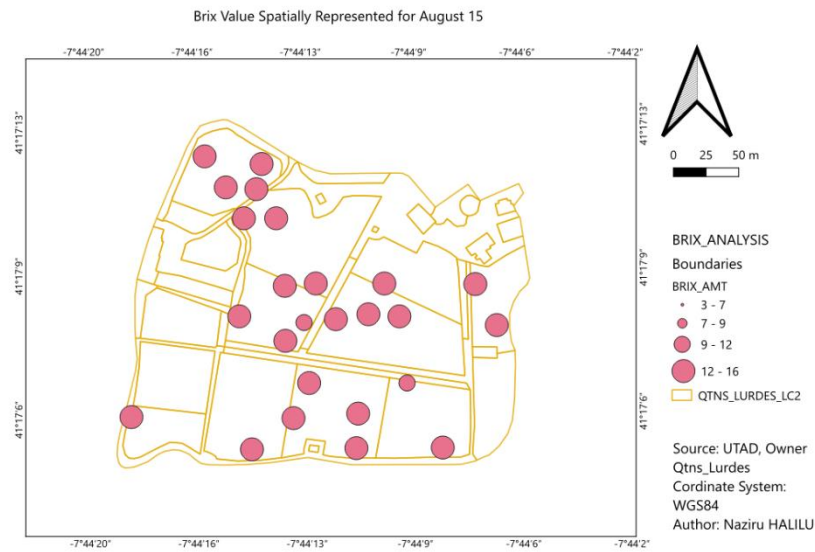


Figure 10: Brix Value spatially represented for Aug 15

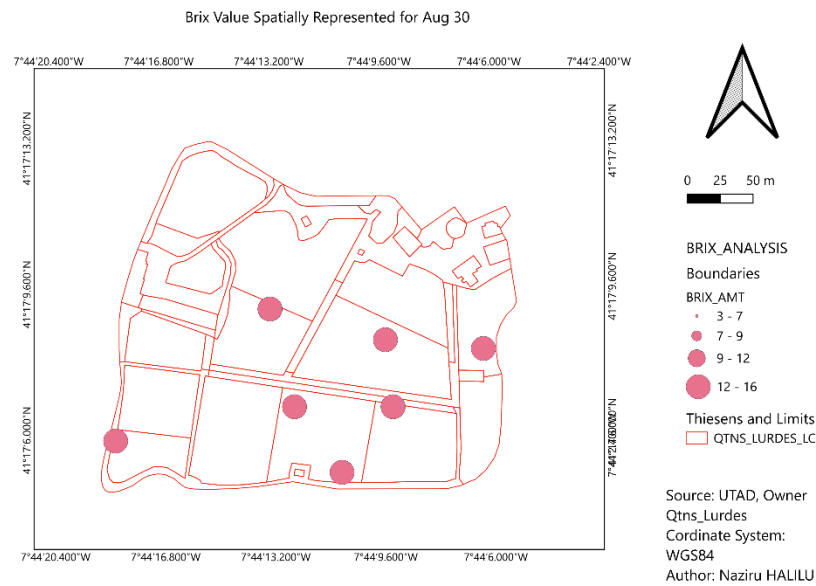


Figure 11: Brix Value spatially represented for Aug 30

4. Statistical Processing and Regression Modeling

Statistical processing and regression modeling are essential tools for transforming sequential Brix observations into predictive management insights. While individual Brix measurements provide valuable snapshots of vineyard maturity at specific dates, regression analysis allows these data points to be mathematically modeled over time, thereby forecasting future sugar accumulation and estimating the optimal harvest date with far greater precision ([Aranha, 2026](#)).

In this vineyard monitoring framework, average Brix values collected across the five sampling dates were plotted against Julian day, a continuous calendar system that represents each day of the year numerically. Using Julian day standardizes temporal analysis and allows for more robust mathematical modeling of biological trends. By associating sugar concentration with time, vineyard managers can move beyond descriptive observation and toward predictive decision-making. The relationship between Brix accumulation and time is not perfectly linear because grape maturation often accelerates during veraison and may slow as grapes approach full ripeness. Therefore, logarithmic regression was selected as an appropriate model because it captures the nonlinear maturation trajectory more accurately than simple linear methods. This approach recognizes that sugar accumulation often follows a biological curve influenced by temperature, sunlight, and physiological thresholds. Through regression modeling, vineyard managers gain the ability to forecast future Brix levels, estimate the likely harvest window, and optimize production schedules. This statistical framework is particularly important for reducing uncertainty in vineyard management, as it transforms periodic sampling into actionable forecasts. In practice, such models help synchronize field operations with winery capacity, ensuring that grapes are harvested neither too early nor too late.

The integration of statistical analysis into vineyard monitoring thus represents a major advancement over purely observational methods, allowing precision viticulture to incorporate predictive intelligence into operational planning.

4.1 Regression Equation

The regression equation developed for this vineyard monitoring study was:

$$y = 65.139 \ln(x) - 335.44 \dots\dots\dots (2)$$

with a coefficient of determination:

$$R^2 = 0.9784$$

This logarithmic equation models the relationship between Julian day (x) and average Brix value (y), offering a highly reliable mathematical representation of grape maturation over time. The use of a logarithmic function reflects the biological reality that sugar accumulation in grapes is not constant but tends to accelerate during veraison before gradually stabilizing near harvest maturity.

The coefficient of determination, $R^2 = 0.9784$, indicates that approximately 97.84% of the variation in Brix values can be explained by the model. This exceptionally high value demonstrates excellent predictive strength and suggests that the observed Brix progression closely follows the regression curve. In practical terms, the model provides vineyard managers with a robust forecasting tool capable of accurately estimating future maturity levels based on current trends.

Such a strong statistical fit is highly valuable because it minimizes uncertainty and increases confidence in harvest planning decisions. Rather than relying solely on subjective judgment or isolated field observations, managers can use this equation to predict when target sugar concentrations are likely to be reached. The equation therefore serves as both a scientific and operational tool, bridging field data collection with strategic management.

4.2 Harvest Forecast

Using the regression model, vineyard managers estimated that Brix values exceeding 25.5–27 °Bx would likely occur near Julian Day 256, corresponding to September 13. This forecast is particularly significant because Brix levels within this range are generally associated with premium harvest maturity and desired alcohol potential for high-quality wine production ([Aranha, 2026](#)).

At approximately 25.5 °Bx, grapes are expected to yield around 14% alcohol, a threshold often associated with optimal sugar maturity for many premium wine styles. Reaching this maturity window ensures that grapes have accumulated sufficient sugars while also allowing for balanced acidity, phenolic development, and flavor complexity.

Forecasting harvest at September 13 provides substantial operational advantages. It allows vineyard managers to anticipate labor needs, prepare harvesting equipment, coordinate winery intake schedules, and strategically sequence parcel harvesting according to maturity gradients. Furthermore, this forecast supports selective harvesting, where higher-performing parcels may be harvested earlier or separately for premium wine production. The predicted harvest date also reduces the risks associated with delayed or premature harvest, such as over-ripening, sugar loss, weather damage, or suboptimal wine composition. By integrating statistical forecasting into vineyard management, the harvest window becomes a calculated decision rather than a reactive one.

4.3 Practical Implications

The practical implications of regression-based harvest forecasting extend far beyond simply identifying a harvest date. One of the most immediate benefits is improved labor scheduling. Vineyard harvest is labor-intensive, and knowing the approximate harvest window in advance allows managers to recruit and allocate workers more efficiently, avoiding shortages or unnecessary idle time.

Winery logistics also benefit substantially from predictive modeling. Processing facilities must be prepared to receive, crush, ferment, and store grapes at the correct time. Accurate forecasts help synchronize grape intake with fermentation tank availability, equipment preparation, and storage planning, thereby reducing operational bottlenecks ([Aranha, 2026](#)).

Parcel prioritization is another major advantage. Because not all vineyard parcels mature uniformly, regression and spatial analysis allow managers to identify which blocks are likely to reach target Brix levels first. This enables differentiated harvest scheduling, where premium parcels can be harvested at peak quality while slower-ripening parcels continue maturing. Selective harvest planning is perhaps the most valuable strategic implication. By combining Brix forecasts with GIS spatial zoning, vineyard managers can separate grapes according to quality, maturity, or intended wine style. This selective approach supports premium product segmentation, better resource allocation, and more precise quality control. Overall, the practical applications of statistical modeling transform vineyard management from generalized scheduling into a precision agriculture system that optimizes timing, quality, and economic efficiency.

5. GIS-Based Monitoring Framework

Geographic Information Systems (GIS) represent a transformative advancement in vineyard monitoring by expanding traditional field-based sampling into a spatially intelligent management framework. While conventional vineyard assessment relies primarily on manual observations and isolated Brix measurements, GIS integrates these observations with precise geographic coordinates, allowing vineyard managers to analyze not only how grapes are maturing over time but also where specific patterns of maturity occur across the landscape. This shift from purely numerical monitoring to spatially explicit analysis is fundamental in precision viticulture because vineyard productivity is rarely uniform. Differences in slope, soil, drainage, aspect, and microclimate create substantial variability between parcels, rows, and even individual vine zones. GIS enables these differences to be visualized, quantified, and strategically managed. The GIS-based monitoring framework begins with the geolocation of sampling points using Global Navigation Satellite Systems (GNSS) and field data collection platforms such as QField. Each sampling location is precisely recorded within the vineyard, ensuring that every Brix reading is linked to an exact geographic position. This georeferenced approach significantly improves data quality because measurements are no longer anonymous values in a spreadsheet; instead, they become spatial records that can be analyzed in relation to terrain, parcel boundaries, and management zones ([Aranha, 2026](#)).

Once sampling points are collected, the attribute table within GIS software is updated with corresponding Brix values for each sampling date. This creates a dynamic spatial database where each point contains not only location information but also temporal maturity data. Such structured data organization is essential for both immediate analysis and long-term vineyard recordkeeping.

The next stage involves the creation of symbolized thematic maps, where Brix values are visually represented using graduated color classes or symbol sizes. This process converts raw numerical data into

intuitive visual patterns, enabling managers to quickly identify areas of low, moderate, or high sugar accumulation across the vineyard. Spatial interpolation techniques are then applied to estimate conditions between sampled points, creating continuous surfaces that reveal broader maturity gradients throughout the property. Finally, parcel zoning translates these analyses into actionable management units. Based on maturity patterns, vineyard blocks can be classified into zones for early harvest, delayed harvest, irrigation support, or selective premium production. This zoning capability is one of GIS's greatest strengths because it transforms spatial analysis into operational strategy. The advantages of GIS-based monitoring are substantial. First, it provides spatial visualization of variability, allowing vineyard managers to clearly understand where productivity differences exist. Second, it supports precision harvest targeting by identifying parcels that reach optimal maturity sooner than others. Third, it improves resource allocation by directing labor, equipment, and inputs only where they are most needed. Finally, GIS creates a historical monitoring archive, enabling year-to-year comparisons and long-term decision support. Together, these capabilities make GIS an indispensable framework for modern vineyard management.

6. Thematic Cartography of Brix Values

Thematic cartography of Brix values is a critical component of precision viticulture because it converts discrete field measurements into visually interpretable maturity landscapes. Rather than relying solely on spreadsheets or numerical summaries, thematic maps present vineyard sugar concentration data spatially, allowing managers to immediately observe how maturation differs between rows, parcels, and topographic zones. This visualization is particularly important in heterogeneous vineyards, where terrain, microclimate, and soil variability can create significant differences in grape development even within small areas.

In practice, Brix measurements collected at georeferenced sampling points are symbolized using graduated color scales or class intervals, often ranging from approximately 3 °Bx to 30 °Bx. Lower Brix values may represent immature zones, while higher values indicate advanced ripening or near-harvest conditions. By applying customized symbology classes, vineyard managers can rapidly interpret spatial maturity patterns and make informed operational decisions. Color gradients, for example, can highlight whether sugar accumulation is concentrated on south-facing slopes, elevated parcels, or specific management blocks.

One of the most important strengths of thematic cartography is its ability to reveal underperforming rows or sectors. Areas consistently showing lower Brix values may indicate poor soil fertility, water stress, shading effects, or disease pressure. Identifying these zones early allows managers to investigate underlying causes and implement corrective actions such as irrigation adjustment, canopy management, or targeted fertilization ([Aranha, 2026](#)).

Thematic maps are also highly effective for detecting delayed maturity zones. Since not all vineyard parcels mature uniformly, certain blocks may lag behind due to elevation, aspect, or local climatic influences. Recognizing these zones prevents premature harvest and supports differentiated timing strategies.

Another major benefit is the ability to compare seasonal progression. By generating thematic maps for multiple sampling dates, vineyard managers can visually track maturation dynamics over time, observing whether slower parcels are catching up or whether variability is widening. This temporal-spatial perspective significantly improves management accuracy. Ultimately, thematic cartography supports selective harvest by enabling vineyard managers to classify zones according to maturity and quality potential. Premium parcels can be harvested separately for superior wine production, while less mature areas can be left longer or allocated differently. In this way, thematic Brix mapping transforms vineyard monitoring from a

generalized process into a precision management system that improves both grape quality and operational efficiency.



Figure 12 : UTAD Enology Experimental July 15_Brix values Spatial distribution



Figure 13 : UTAD Enology Experimental July 30_Brix values Spatial distribution



Figure 14: UTAD Enology Experimental August 6_Brix values Spatial distribution



Figure 15: UTAD Enology Experimental August 15_Brix values Spatial distribution



Figure 16: UTAD Enology Experimental August 30_Brix values Spatial distribution

7. Geostatistical Allocation: Thiessen Polygons

Thiessen polygons, also known as Voronoi diagrams, provide one of the simplest and most practical geostatistical allocation methods for translating discrete vineyard sampling points into spatial management zones. This deterministic technique assigns each location within the vineyard to the nearest sampled point, creating polygons where every area inside a polygon is assumed to be best represented by the Brix value of its corresponding sample. In vineyard monitoring, Thiessen polygons are particularly useful when managers need a rapid and straightforward method for spatially allocating maturity values across a property.

The core principle of Thiessen polygons is proximity. Each sampling point becomes the center of influence for its surrounding territory, and boundaries are drawn equidistant between neighboring points. This creates a complete tessellation of the vineyard, ensuring that every part of the landscape is assigned to one sampled observation. For vineyard managers, this method is valuable because it transforms point-based Brix data into area-based maturity zones without requiring complex interpolation modeling.

One of the greatest benefits of Thiessen polygons is their speed and simplicity. They are computationally efficient, easy to generate in GIS platforms, and highly intuitive to interpret. For vineyards with moderate sampling density, Thiessen polygons provide a practical first approximation of spatial variability and are especially useful for early-stage zoning or operational planning. Thiessen polygons are beneficial when sample density is sufficient but not extensive enough for more advanced interpolation methods. They provide a deterministic representation that can support management decisions such as dividing harvest blocks, comparing zones, or identifying approximate maturity sectors. However, Thiessen polygons also have important limitations. Their most notable weakness is the creation of sharp artificial boundaries between polygons. In reality, vineyard conditions such as soil moisture, sugar accumulation, and vine vigor change gradually across space rather than abruptly. Thiessen polygons may therefore oversimplify natural transitions by imposing rigid divisions where none truly exist.

Another limitation is the absence of gradual transitions or weighted influence from surrounding points. Unlike interpolation methods such as IDW or kriging, Thiessen polygons assume that the nearest sample fully represents an area regardless of nearby values, potentially reducing realism in highly variable landscapes. Despite these constraints, Thiessen polygons remain a highly useful geostatistical tool in vineyard management. They provide a fast, deterministic, and visually accessible method for allocating maturity data, particularly when used as an intermediate step before more sophisticated interpolation. In practical vineyard operations, they serve as an efficient bridge between field sampling and spatial decision-making, supporting parcel organization, harvest planning, and preliminary zoning.

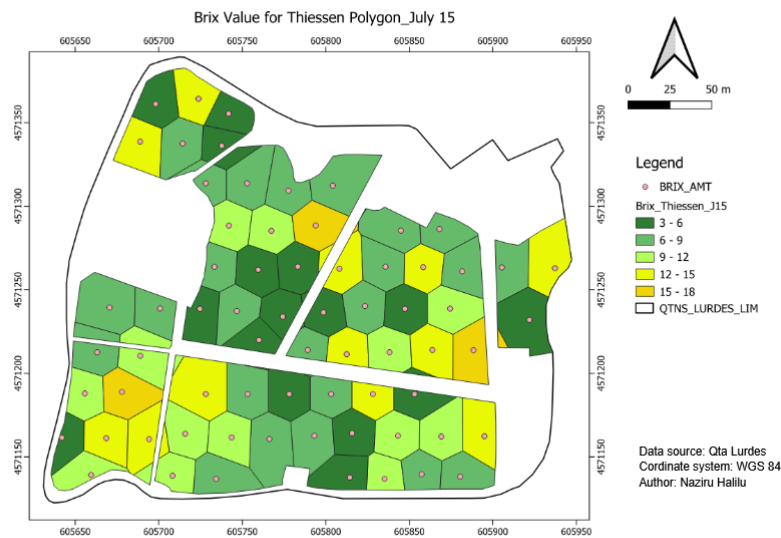


Figure 17: July 15 Brix Values in Thiesen and Voronoi Polygons

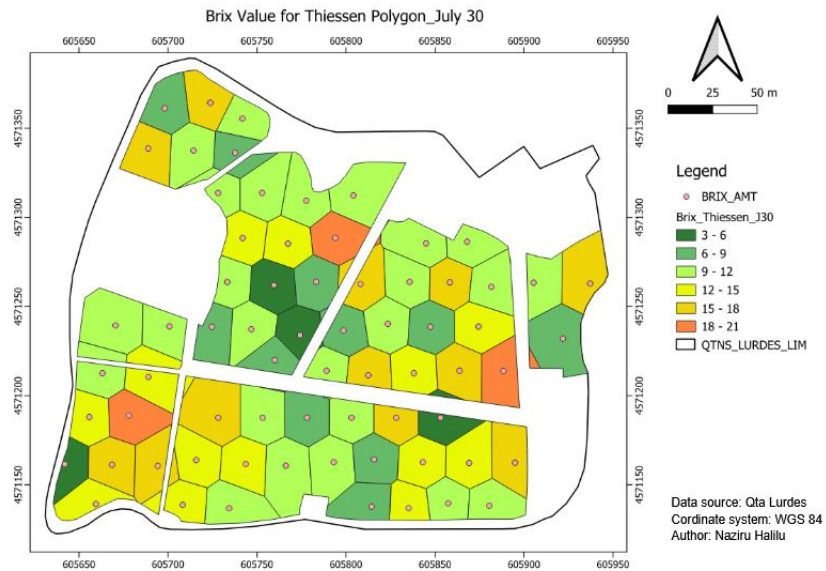


Figure 18: July 30 Brix Values in Thiesen and Voronoi Polygons

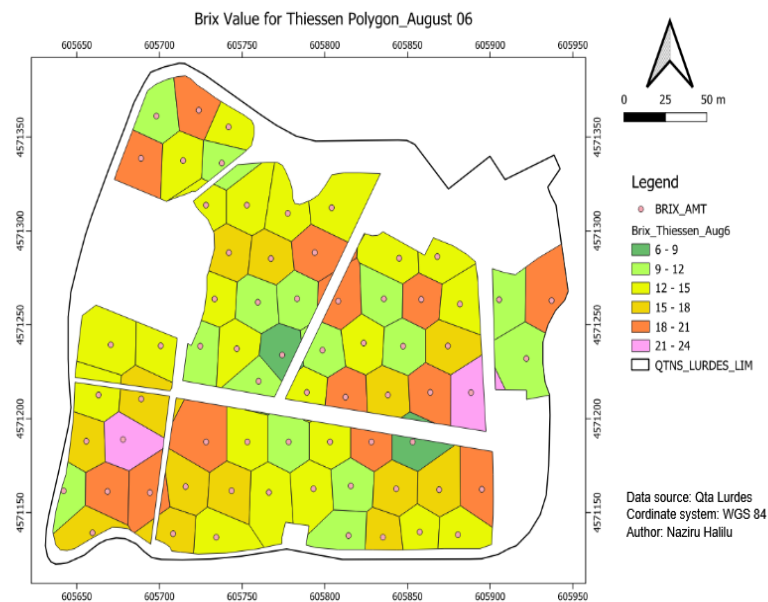


Figure 19: August 06 Brix Values in Thiesen and Voronoi Polygons

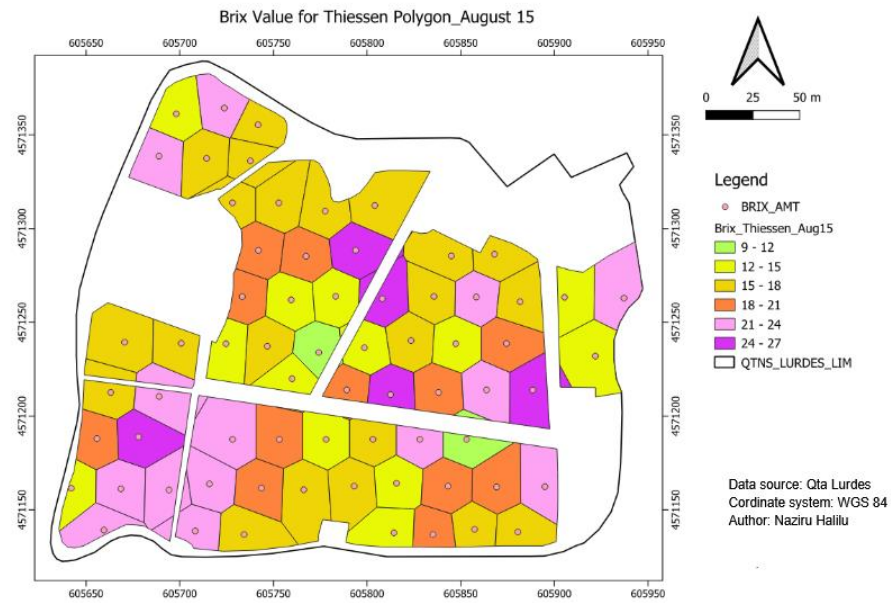


Figure 17: August 15 Brix Values in Thiesen and Voronoi Polygons

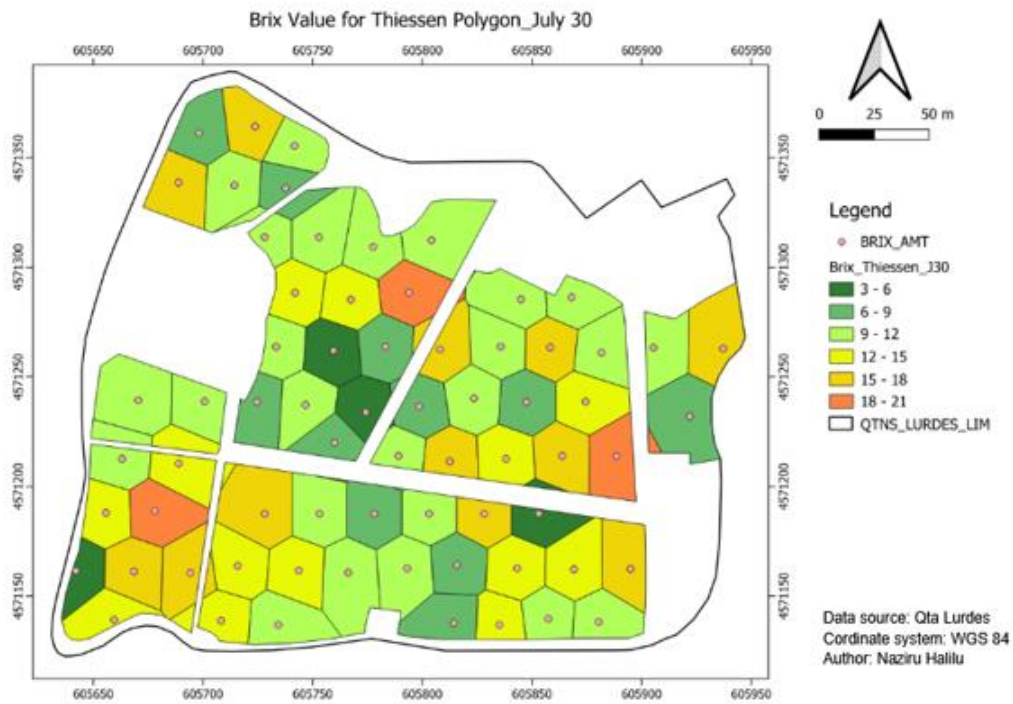


Figure 17: August 30 Brix Values in Thiesen and Voronoi Polygons

8. Deterministic Interpolation: Inverse Distance Weighted (IDW)

Inverse Distance Weighted (IDW) interpolation is one of the most practical and widely applied deterministic geostatistical methods used in precision viticulture for transforming discrete sampling points into continuous spatial surfaces. In vineyard monitoring, field-collected Brix values are inherently point-specific, meaning they only describe sugar concentration at exact sampled locations. However, vineyard managers require a broader understanding of how maturity varies across the entire property, including unsampled areas. IDW addresses this challenge by estimating values between sampled points based on the principle that locations closer to one another are more similar than those farther apart. This assumption aligns well with vineyard environments, where nearby vines often share similar soil, topographic, and microclimatic conditions. The mathematical foundation of IDW is based on assigning greater influence to nearby observations and progressively less influence to distant points. Each estimated location receives a weighted average of surrounding sample values, with weights decreasing as distance increases. As a result, IDW creates a smooth, continuous surface of predicted Brix values across the vineyard landscape. This makes it particularly effective for representing gradual biological transitions, such as sugar accumulation, that naturally occur over space. One of the primary strengths of IDW is its ability to produce continuous maturity surfaces, which provide vineyard managers with a much more realistic representation of grape development than discrete points or rigid polygon boundaries. Instead of abrupt divisions, IDW reflects gradual spatial variability, allowing managers to see subtle maturity gradients from one parcel to another. This is especially important in heterogeneous vineyards, where changes in elevation, slope, or soil moisture may progressively influence grape ripening. IDW is also highly suitable for vineyard zoning because it translates complex sampling datasets into operationally meaningful maturity zones. By identifying areas of similar Brix concentration, managers can divide the vineyard into sectors based on expected harvest readiness or wine quality potential. This zoning capability supports site-specific management and improves production efficiency. In practical terms, IDW has several valuable applications. It can identify high-Brix premium sectors where grapes may be best suited for top-tier wine production, allowing for selective harvesting and differentiated vinification. It can also locate low-performance sectors where delayed maturity, poor vigor, or environmental constraints may require additional management interventions such as irrigation or fertilization. Furthermore, IDW supports precision harvest scheduling by helping managers determine which zones should be harvested first and which may benefit from additional ripening time. Overall, IDW represents a powerful bridge between field sampling and precision agriculture by converting localized observations into vineyard-wide decision support. Its balance of simplicity, realism, and operational value makes it one of the most effective interpolation tools for vineyard maturity assessment.

9. Remote Sensing Integration

Remote sensing has emerged as a transformative component of modern vineyard assessment, significantly enhancing monitoring efficiency by reducing field labor requirements while simultaneously increasing spatial precision and analytical depth. Traditional vineyard monitoring methods, although reliable, are labor-intensive and often constrained by limited sampling density. Remote sensing overcomes these limitations by enabling large-scale, high-resolution observation of vineyard conditions through aerial and spectral technologies. At Quinta de Nossa Senhora de Lurdes, the integration of drone-based remote sensing

complements field sampling by providing detailed geospatial information that supports precision management. The technological foundation of this approach includes advanced drone systems such as the DJI Phantom 4 RTK and Mavic 3 Thermal. The DJI Phantom 4 RTK provides high-precision georeferenced imagery through Real-Time Kinematic positioning, allowing for centimeter-level mapping accuracy. This precision is essential for vineyard zoning, row analysis, and integration with GIS databases. The Mavic 3 Thermal adds another layer of functionality by capturing thermal infrared data, enabling vineyard managers to detect temperature anomalies, plant stress, and irrigation irregularities.

These drones collect multiple forms of imagery, including RGB (Red, Green, Blue), Near Infrared (NIR), and Red Edge spectral bands. RGB imagery provides true-color visual representations of vineyard conditions, while NIR and Red Edge data are essential for vegetation health analysis, particularly NDVI generation. By capturing multiple spectral bands, remote sensing allows for a more comprehensive understanding of vine vigor, canopy health, and stress patterns than visual observation alone.

Agisoft photogrammetry software is then used to process aerial images into geospatially accurate outputs. Through image stitching, orthorectification, and mosaicking, raw drone photographs are transformed into orthomosaics high-resolution spatial maps that provide detailed vineyard overviews. Additional outputs include NDVI maps, which estimate vegetation vigor; thermal stress maps, which reveal heat or water stress; and land cover precision maps, which support detailed parcel characterization. These outputs provide vineyard managers with powerful tools for operational planning. Orthomosaics improve spatial orientation and parcel management, NDVI maps support productivity analysis, thermal maps identify stress zones, and precision land cover maps enhance long-term vineyard characterization. Collectively, remote sensing transforms vineyard monitoring from periodic manual inspection into a scalable, data-rich, precision management system.

10. NDVI for Vineyard Productivity

The Normalized Difference Vegetation Index (NDVI) is one of the most valuable remote sensing tools for assessing vineyard productivity because it provides a quantitative measure of vegetation vigor based on spectral reflectance properties. Healthy vegetation reflects near-infrared light strongly while absorbing red light for photosynthesis. NDVI exploits this difference to estimate plant health, vigor, and biomass distribution across agricultural landscapes.

The NDVI formula is expressed as:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

This equation produces values typically ranging from -1 to +1, where higher positive values indicate healthier, more vigorous vegetation. In vineyard systems, NDVI is particularly useful because vine vigor is closely linked to canopy development, photosynthetic activity, and, indirectly, grape maturation potential. One of the major benefits of NDVI is its ability to detect vigor variability across vineyard parcels. Different sections of the vineyard may exhibit varying growth rates due to differences in soil fertility, moisture availability, slope, or disease presence. NDVI maps allow these variations to be visualized spatially, enabling managers to identify both strong and weak sectors. NDVI is also highly effective for water stress assessment. Since water stress influences photosynthetic efficiency and canopy reflectance, lower NDVI values may indicate drought stress or irrigation deficiencies. This allows managers to target irrigation interventions more precisely rather than applying water uniformly across the vineyard. Fertility management also benefits from NDVI because areas of reduced vigor may reflect nutrient limitations. By

identifying these zones, fertilization can be targeted more efficiently, improving both cost-effectiveness and sustainability. Perhaps most importantly, NDVI has shown predictive potential for Brix estimation. In this study, regression analysis between NDVI and Brix produced an R^2 value of 0.8795, indicating a strong predictive relationship. This suggests that NDVI can serve as a remote proxy for grape maturity, allowing vineyard managers to estimate sugar accumulation patterns even without exhaustive physical sampling. Thus, NDVI not only enhances vineyard monitoring efficiency but also strengthens predictive viticulture by linking canopy vigor to harvest quality.

The pivot table reveals a strong relationship between NDVI-based vegetation vigor and grape maturation performance across the vineyard. Parcels classified under very high NDVI consistently showed the highest Brix values throughout the monitoring period, indicating that healthier and more vigorous vines generally accumulated sugars more rapidly and reached harvest maturity earlier than lower-vigor zones. These parcels are particularly important for premium wine production because their superior canopy performance likely supported better photosynthetic efficiency, resulting in enhanced sugar concentration and improved harvest potential. Moderate NDVI parcels represented the largest proportion of the vineyard and demonstrated balanced, stable maturation trends. Their Brix progression suggests that these areas form the productive core of the vineyard, contributing consistently to standard harvest quality without requiring extreme management interventions. High NDVI sectors also performed well, showing strong maturity acceleration and favorable harvest outcomes, though not always reaching the elite levels of the highest-vigor zones.

In contrast, parcels within the very low and low NDVI classes exhibited delayed ripening and lower final Brix values, suggesting reduced productivity potential. These areas may be experiencing environmental or agronomic constraints such as poor soil fertility, inadequate water availability, shading, or localized stress. Such zones should be prioritized for corrective interventions including targeted irrigation, nutrient management, or canopy adjustment to improve future performance.

Overall, the table demonstrates that NDVI is a valuable predictor of vineyard productivity and can effectively support precision agriculture decisions. By linking canopy vigor with temporal sugar accumulation, vineyard managers can identify premium sectors for selective harvesting, detect underperforming zones for intervention, and optimize resource allocation across the property. This relationship reinforces the importance of integrating NDVI, Brix monitoring, and GIS analysis into a comprehensive precision viticulture framework.

Table 2: NDVI Class and Average Vineyard Performance

NDVI Class	Parcel Count	Avg Jul 15	Avg Jul 30	Avg Aug 06	Avg Aug 15	Avg Aug 30	Avg Brix Increase (Jul15–Aug30)	Avg Daily Maturity Rate	Interpretation
Very Low (<0.15)	8	8.13	10.88	13.50	17.00	19.13	11.00	0.24 °Bx/day	Weak vigor, delayed ripening, likely water or nutrient stress
Low (0.15–0.25)	11	8.09	10.91	13.82	17.18	19.36	11.27	0.25 °Bx/day	Moderate underperformance, selective intervention needed
Moderate (0.25–0.35)	30	9.77	12.67	15.63	19.17	21.13	11.37	0.25 °Bx/day	Balanced productivity, stable maturation
High (0.35–0.45)	15	8.73	11.67	14.87	18.53	20.73	12.00	0.26 °Bx/day	Strong vigor, good maturation acceleration
Very High (>0.45)	6	12.17	15.00	18.00	22.33	23.83	11.66	0.26 °Bx/day	Premium vigor, early maturity, harvest priority

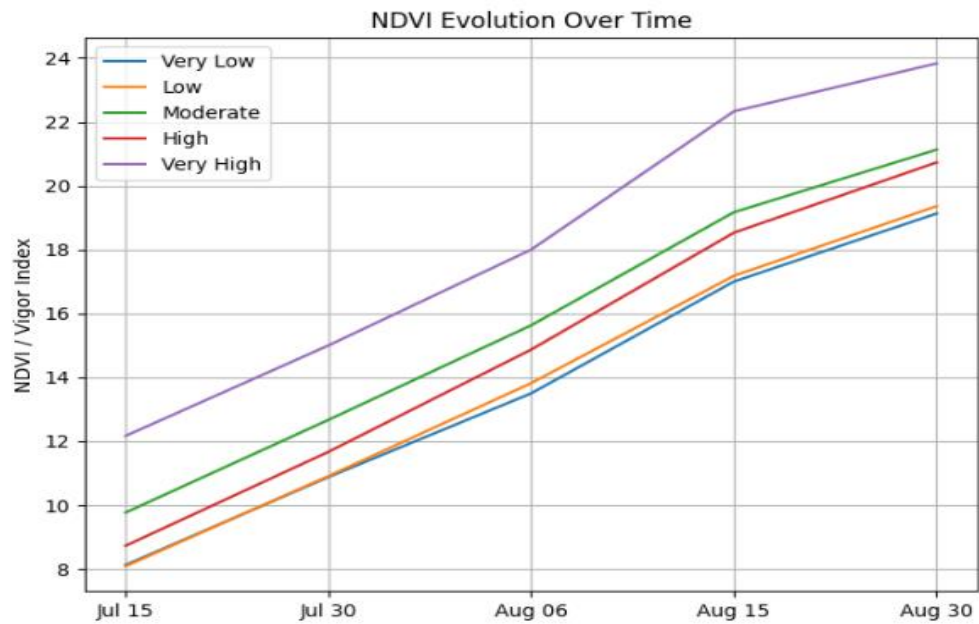


Fig 18 NDVI Evolution over time

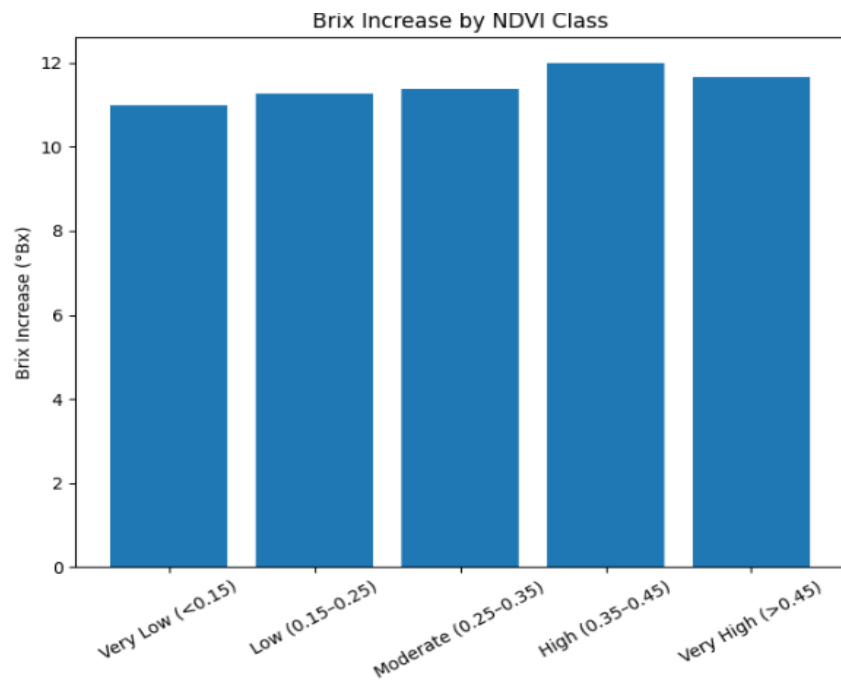


Fig 19: Brix Increment in Relation to NDVI

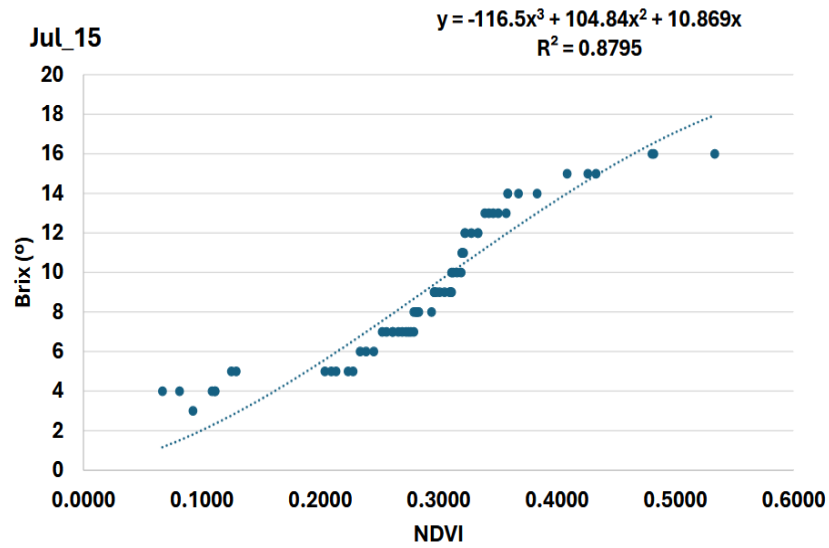


Figure 20 Adjusting the Regresiom Model

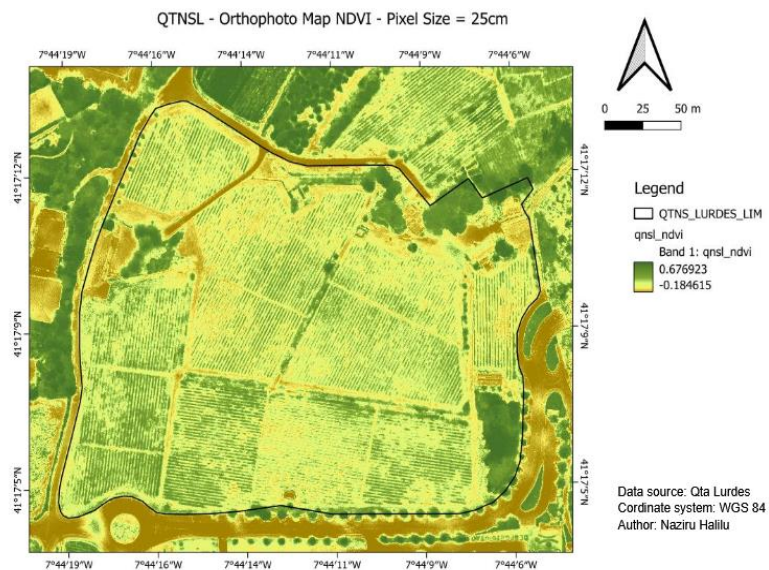


Figure 21 Normalised difference Vegetation Index

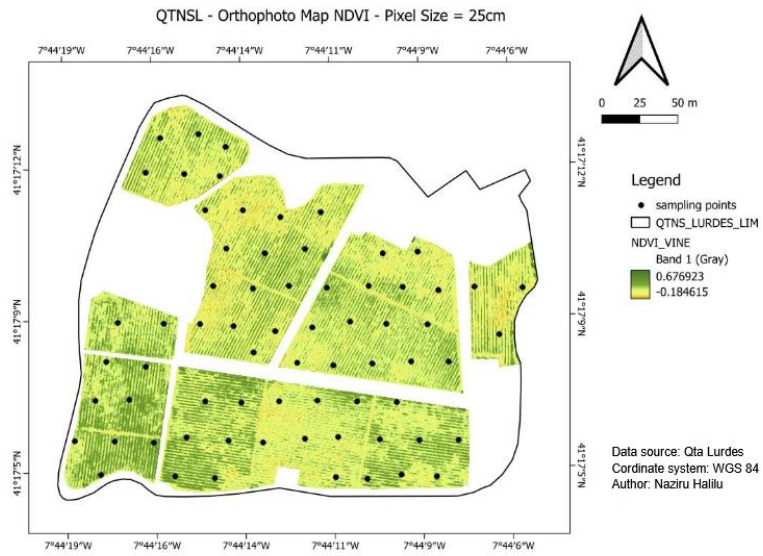


Figure 22 Normalised difference Vegetation Index Clipped

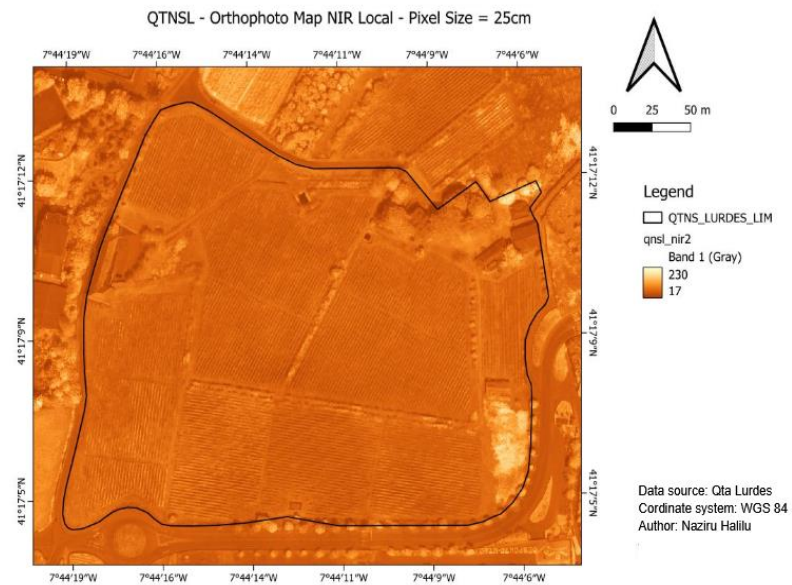


Figure 23 Ortho Photo Map of NIR

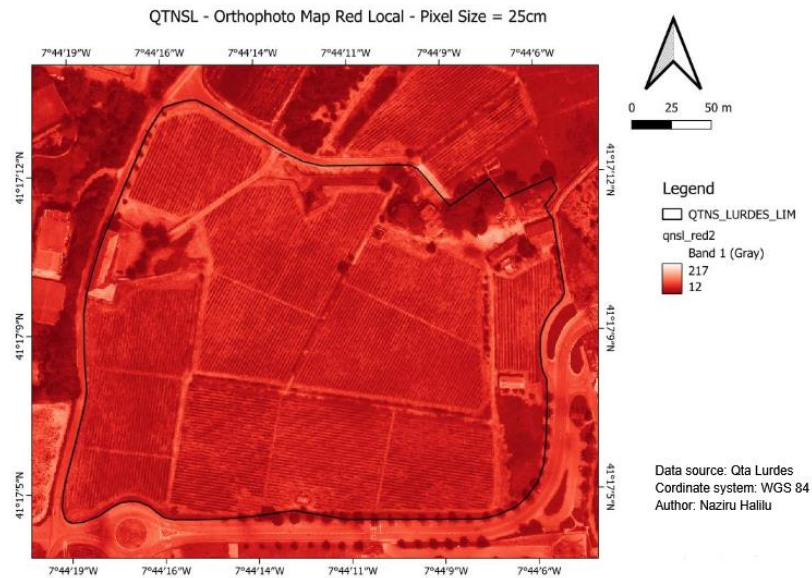


Figure 24 Qtns Lurdes NIR

11. Precision Agriculture Decision Support

The integration of Brix measurements, GIS spatial analysis, and NDVI remote sensing creates a highly sophisticated precision agriculture decision support system capable of transforming vineyard management into a site-specific, data-driven process. Rather than managing the vineyard as a uniform entity, this integrated framework recognizes spatial heterogeneity and supports tailored interventions for individual parcels or zones. From a harvest perspective, this system enables the identification of early harvest zones where grapes have reached maturity sooner due to favorable slope, aspect, or vigor conditions. These areas can be prioritized to maximize sugar concentration and wine quality. Premium wine parcels with optimal Brix and vigor profiles can be harvested selectively for superior production lines, while delayed parcels can remain longer on the vine to achieve target maturity. Differential vinification strategies can also be implemented, where grapes from different maturity zones are processed separately according to quality goals. Management decisions are equally enhanced. Irrigation targeting becomes more efficient when low-vigor or water-stressed zones are spatially identified. Fertilizer zoning allows nutrient inputs to be focused only where deficiencies exist, reducing waste and environmental impact. Disease scouting can be prioritized in zones showing abnormal vigor or thermal stress, improving pest and disease response efficiency. Yield optimization becomes more strategic because vineyard managers can align resource allocation with productivity potential. This integrated decision support framework significantly improves operational precision, economic returns, and sustainability outcomes.

12. Operational Workflow Summary

The operational workflow for precision vineyard monitoring follows a structured multi-phase framework that integrates field sampling, GIS processing, remote sensing, and strategic decision-making into a coherent management system. Phase 1 begins with field sampling, where grape berries are collected from designated plots, sugar concentration is measured using a refractometer, and geographic coordinates are captured using GNSS-enabled devices. This phase establishes the foundational dataset for all subsequent analyses. Phase 2 involves GIS processing, where field data are entered into spatial databases, mapped using thematic cartography, and analyzed through interpolation techniques such as Thiessen polygons or IDW. This transforms raw field observations into vineyard-wide spatial intelligence.

Phase 3 incorporates remote sensing through drone flights equipped with RGB, NIR, and thermal sensors. NDVI extraction and parcel comparison further expand understanding of vigor, stress, and maturity patterns. Finally, Phase 4 focuses on decision-making, where all collected information is synthesized to forecast harvest dates, define quality zones, and optimize resource allocation. This final stage transforms technical analysis into practical vineyard strategy. Together, these phases form a comprehensive precision viticulture workflow that maximizes grape quality, improves operational efficiency, and supports sustainable vineyard management.

13. Economic and Technical Benefits

The adoption of precision viticulture technologies within vineyard management offers substantial economic and technical benefits that extend beyond simple productivity gains. By integrating traditional Brix sampling with GIS, geostatistics, and remote sensing, vineyard operations become more efficient, strategic, and profitable. These benefits are particularly significant in premium viticultural systems such as Quinta de Nossa Senhora de Lurdes, where small improvements in grape quality or harvest timing can produce substantial financial returns. From an economic perspective, one of the most immediate advantages is the reduction of unnecessary harvest delays. Traditional harvest decisions based solely on generalized observations may lead to grapes remaining on the vine beyond optimal maturity or being harvested prematurely. Both scenarios can reduce wine quality, compromise alcohol targets, or expose grapes to climatic risks such as rainfall or disease outbreaks. Precision monitoring minimizes this uncertainty by providing accurate maturity forecasts, enabling harvest to occur at the ideal time. Better winery scheduling is another critical economic benefit. Harvest timing affects crusher utilization, fermentation tank availability, labor deployment, and supply chain logistics. Accurate maturity predictions allow wineries to coordinate grape intake more efficiently, reducing operational bottlenecks and maximizing processing capacity. This synchronization can significantly improve workflow and reduce labor inefficiencies. Improved grape quality premiums represent one of the most financially valuable outcomes. By identifying high-performing parcels with superior Brix or vigor profiles, vineyard managers can selectively harvest premium grapes for higher-value wine categories. Differential vinification can create product segmentation, increasing market competitiveness and profitability. Precision viticulture also reduces blanket fertilization and other uniform input applications. Instead of applying fertilizers or irrigation equally across all parcels, managers can target only underperforming or stressed zones. This targeted approach lowers input costs, reduces environmental impact, and enhances sustainability. From a technical standpoint, precision systems provide far higher spatial precision than conventional methods. Managers gain detailed insight into within-vineyard variability, enabling site-specific management strategies. Better prediction accuracy also enhances operational planning by reducing uncertainty around maturity progression and harvest timing.

Historical database creation is another major technical advantage. Over time, GIS and remote sensing systems build temporal records that reveal recurring patterns, seasonal variability, and long-term vineyard trends. These datasets improve strategic planning and research potential. Finally, the methodology is scalable, meaning that once systems are established, they can be expanded to larger vineyards, additional crops, or broader regional applications with relatively low marginal costs. Thus, the economic and technical advantages of precision viticulture create both immediate and long-term value.

14. Constraints and Limitations

Despite its substantial advantages, precision viticulture also presents several constraints and limitations that must be carefully considered before implementation. These challenges are not necessarily barriers but rather operational realities that influence adoption speed, investment requirements, and system effectiveness.

One of the primary limitations is the initial GIS training requirement. Geographic Information Systems are powerful tools, but their effective use requires technical knowledge in data management, spatial analysis, interpolation, and cartographic interpretation. Vineyard managers or technicians unfamiliar with GIS may require substantial training before fully benefiting from these systems.

Drone acquisition costs also represent a significant financial constraint, particularly for small-scale producers. High-quality drones such as the DJI Phantom 4 RTK or thermal-enabled systems involve considerable upfront investment, along with software costs for photogrammetry and image analysis. Although these technologies often produce long-term returns, the initial expense may discourage adoption. Calibration is another critical technical challenge. Remote sensing outputs such as NDVI are highly valuable, but without regular field validation through Brix sampling or ground truthing, predictive accuracy may decline. This means traditional fieldwork remains essential, particularly during model development stages. Sampling intensity requirements also influence reliability. Sparse sampling can reduce interpolation accuracy, while overly dense sampling increases labor demands. Determining the optimal balance between cost and precision is essential for operational efficiency. Weather constraints can significantly affect drone operations. Wind, rain, cloud cover, and lighting conditions may reduce image quality or prevent flights altogether, limiting data acquisition windows. Despite these limitations, the long-term gains of precision viticulture generally exceed implementation complexity. Once systems are operational, benefits in efficiency, quality, and sustainability often outweigh the initial barriers, making these technologies highly valuable strategic investments.

15. Comparative Framework

A comparative evaluation of vineyard monitoring methods reveals that no single approach is universally superior; rather, each method offers distinct strengths and weaknesses depending on operational goals, budget, and technical capacity. Traditional sampling remains the most accessible method due to its low cost and minimal technological requirements. Refractometers are inexpensive and reliable, making them suitable for baseline maturity monitoring. However, this method is highly labor-intensive and spatially limited, as it only represents sampled points. GIS mapping offers substantial improvements through spatial precision, enabling managers to visualize variability and make site-specific decisions. However, GIS requires technical skills and software expertise, which may present barriers for less technologically prepared operations. Thiessen polygons provide fast deterministic allocation and are easy to implement. Their simplicity makes them useful for preliminary zoning, but their abrupt boundaries often oversimplify natural vineyard variability. IDW interpolation offers continuous surfaces and more realistic maturity

gradients, making it particularly effective for vineyard zoning and harvest planning. However, its accuracy is sensitive to sample spacing and distribution. NDVI delivers large-scale efficiency by enabling rapid, broad-scale assessment of vine vigor and stress. It significantly reduces labor demands but requires calibration with field data to ensure reliable maturity prediction. Ultimately, combining these methods often yields the strongest results, with traditional sampling providing calibration, GIS enabling spatial analysis, IDW supporting interpolation, and NDVI expanding large-scale predictive capacity.

16. Strategic Recommendations

To maximize the effectiveness of precision viticulture systems, several strategic recommendations should guide implementation and operational planning. First, traditional Brix sampling should be maintained as the calibration foundation for all predictive systems. While advanced technologies provide scalability, field measurements remain essential for validation. Second, GNSS-tagged sample density should be expanded to improve spatial interpolation quality and increase predictive reliability. More sampling points generally enhance zoning precision. Third, IDW should be prioritized for operational harvest zoning because it balances simplicity with realistic continuous surface generation. Fourth, NDVI should be integrated with periodic field validation to ensure that spectral indicators remain accurately linked to grape maturity and productivity. Fifth, annual temporal GIS databases should be developed to build historical records, improve trend analysis, and support long-term strategic planning. Finally, drone thermal imagery should be increasingly used for water stress detection, irrigation optimization, and early anomaly identification.

These recommendations collectively create a resilient and scalable precision viticulture framework.

17. Future Directions

The future of vineyard management is likely to be shaped by increasingly sophisticated digital technologies that move beyond precision agriculture into intelligent autonomous systems.

AI-driven harvest prediction represents one of the most promising developments, where machine learning algorithms could integrate weather, Brix, NDVI, and historical datasets to generate highly accurate dynamic forecasts. Machine learning NDVI-Brix models could further improve remote maturity prediction by refining spectral relationships across seasons and varieties. Automated robotic scouting systems may reduce labor dependency by using autonomous vehicles or drones for continuous parcel surveillance. Disease early warning systems could combine thermal imaging, spectral anomalies, and predictive analytics to detect pest or pathogen outbreaks before visible symptoms appear. Perhaps the most advanced future direction is the development of vineyard digital twins virtual replicas of real vineyards that continuously integrate environmental, physiological, and operational data for simulation-based decision-making. These technologies could fundamentally redefine vineyard management in coming decades.

18. Conclusion

Quinta de Nossa Senhora de Lurdes provides a compelling example of how precision viticulture can transform vineyard management from generalized agricultural practice into a sophisticated, data-driven decision system. While traditional refractometer-based Brix sampling remains foundational for assessing grape maturity, the integration of GIS, geostatistics, and remote sensing technologies dramatically expands its strategic value by converting isolated field observations into comprehensive whole-property intelligence. The vineyard's observed maturation curve suggests an optimal harvest window near September 13, when Brix values are projected to achieve premium alcohol thresholds suitable for high-quality wine production. Through GIS mapping and geostatistical tools such as Thiessen polygons and IDW, vineyard managers can spatially differentiate parcel performance, while NDVI and drone technologies further enhance productivity

assessment through scalable remote sensing. This integrated framework enables selective harvesting, optimized resource allocation, improved economic returns, and greater environmental sustainability. By combining field precision with digital intelligence, Quinta de Nossa Senhora de Lurdes demonstrates a model of technologically advanced viticulture that is highly relevant not only within the Douro region but also across broader European wine systems. Its approach illustrates how precision agriculture can support sustainable, profitable, and scientifically informed vineyard management in the modern era.

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QGIS Geographic Information System Documentation.

DJI Precision Agriculture Imaging Systems.

NDVI Remote Sensing Methodologies.

Appendices

Appendix A: Key Formulae

- $1\text{ }^{\circ}\text{Bx} \approx 0.55\text{ }^{\circ}\text{Alcohol}$
- Regression: $y = 65.139\ln(x) - 335.44$
- $\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$